

Aaryan Lath

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Aeronautical and Astronautical Engineering (GPA: 3.74/4.0) Expected December 2026

Specialization in Aerodynamics and Systems Design

Honors: Dean's List and Semester Honors for every semester, **AAE Ambassador**

SKILLS

Core Tools: CAD, NX, CREO, Assembly, GD&T, Teamcenter, Ansys, Product Design, JIRA, CAM, SAP

Programming & Analysis: Python, MATLAB, Microsoft Office Suite, CFD, FEA, HTML, Java

Soft-skills: Analytical, Communicator, Detail-oriented, Fast-paced learner, Inquisitive

Awards: 2nd place in World Robot Olympiad Junior High Nationals

PROFESSIONAL EXPERIENCE

Zucrow High Speed Compressors Lab

West Lafayette, IN

Undergraduate Research Assistant

Aug 2025 – Present

- Researching into the inlet vortex distortion phenomena and their effects on turbomachinery performance.
- Helped assemble and calibrate the test cell's fan-rig using laser alignment, reducing time taken by **15%**.

Siemens USA

Grand Prairie, TX

Systems Engineering Intern

May 2025 – Aug 2025

- Designed custom enclosures for panelboards using **CREO** and executed ECNs in **SAP**.
- Streamlined **100+** switchboard neutral assemblies configurations by resolving edge case designs.
- Automated BOM generation with **Python**, cutting manual processing time by **~30%** for **10+** orders/day.

Purdue School of Aeronautics and Astronautics

West Lafayette, IN

Undergraduate Teaching Assistant

Jan 2025 – Present

- Led study sessions to teach **50+** students, core **AAE 251** (Aircraft and Spacecraft Design) course material.
- Provided personalized support to students with concepts and **MATLAB** troubleshooting.
- Guided students in understanding key design principles for the aircraft and spacecraft design project.

Resilient Extraterrestrial Habitat Engineering

West Lafayette, IN

Undergraduate Research Assistant - Systems Engineer

May 2024 – May 2025

- Analyzed **18** potential disruptions scenarios in a lunar habitat through conducting **Simulink** simulations.
- Performed trade studies to evaluate safety controls, optimizing habitat resilience while minimizing costs through **Equivalent System Mass** analysis.
- Executed **FEA** simulations and tests to validate **vibration isolation** devices under various conditions.

PROJECTS AND INVOLVEMENTS

SAE Aero Design

West Lafayette, IN

Chief Engineer/ Program Manager

May 2025 – Present

- Authored a **~75 page** RC aircraft design guide to accelerate onboarding.
- Documented past failures from the SAE Aero Design Regular Class to avoid recurring errors.
- Led technical development for **3** sub teams by reviewing trade studies and ensured milestones are met.

Purdue Space Program – A SEDS Chapter

West Lafayette, IN

Technical Director

Dec 2024 – May 2025

- Optimized the clubs backend operations and handled 9 sub-teams for smooth project completion.

Satellites Deputy Systems Director

Feb 2024 – Nov 2024

- Improved project progress with **~15** engineers and other sub-teams ensuring alignment with project goals.
- Formulated, reviewed, and refined **200** project and system-level requirements for Boiler Bus.
- **Manufactured** the satellite frame end plates using **CAM** and walls through **laser cutting**.

Purdue Aerial Robotics Team

West Lafayette, IN

Competition Airframe

Oct 2023 – May 2024

- Performed analysis using XFLR5 to select optimal airfoils, improving lift-to-drag efficiency by **~10%**.
- Applied engineering solutions to resolve manufacturing issue in **molds, carbon fiber**, and internal structures.
- Engineered aircraft parts for the AUVSI SUAS competition through **NX** and Teamcenter with **~30** engineers.